

Document Understanding with SPARK

Analyse AASB ED SR1 submission letters with an LLM flow - then scale it across many files and highlight the evidence (slides 13-20).

Activity is designed by Su Nguyen; Detailed guide is drafted by Claude Cowork.

Before you start

You need a Google AI (Gemini) API key - the same one you created in the Week 5 SPARK activity from Google AI Studio. SPARK runs in your browser at the staging link your instructor shares. Open Settings (gear, top-right), choose Google Gemini as the AI Provider, paste your key, and click Save Settings before building any flow. Never share or screenshot your key. The submission letters used here (edsr1_nab.pdf, edsr1_aba.pdf, edsr1_coba.pdf) are in the Week 6 > data > aasb folder.

Tool	SPARK - Studio for Prototyping Agent Reasoning & Knowledge
Model	Google Gemini (gemini-2.5-flash)
Input	AASB ED SR1 comment letters from the banking industry (PDF)
What you build	PDF File -> LLM -> Display, then an Iterator for many files, then a Text Highlighter
Outcome	A per-letter sentiment summary, a batch run across 3 letters, and a PDF with the key evidence highlighted

Three companion screen recordings (Week6_SPARK_SingleLetter.gif, Week6_SPARK_Iterator.gif, Week6_SPARK_TextHighlighter.gif) show every click in sequence.

Step 1 - Understand the data: AASB ED SR1 submission letters (Slides 13, 15-16)

The Australian Accounting Standards Board (AASB) released Exposure Draft ED SR1, which proposes Australia's first climate-related financial disclosure standards (the draft ASRS Standards, built on the international IFRS S1 and S2 baseline). Stakeholders responded with public comment letters.

In this activity you act as an analyst who has to read those letters quickly. Instead of reading each one end to end, you build a small AI flow that summarises each organisation's sentiment towards the draft.

The letters you will use (already in Week 6 > data > aasb):

- edsr1_nab.pdf - National Australia Bank (a major retail bank)
- edsr1_aba.pdf - Australian Banking Association (the banks' peak body)

- eds1_coba.pdf - Customer Owned Banking Association (mutual banks and credit unions)

Tip: the full list of submitters is on the AASB website. The demo deliberately picks a few letters from the banking industry so you can compare related organisations.

Step 2 - Load the PDF Q&A example as a starting point (Slides 14, 17)

Open SPARK and go to the Gallery. The quickest way to a document-analysis flow is to start from a ready-made example and adapt it.

1. On the Gallery tab, find the PDF Q&A card ("Upload a PDF, ask a question, and get a direct answer").
2. Click Load Example. If asked to clear the current canvas, click Confirm.
3. The Canvas opens with four nodes: Load PDF, My Question, Answer Generator, and Display Answer.

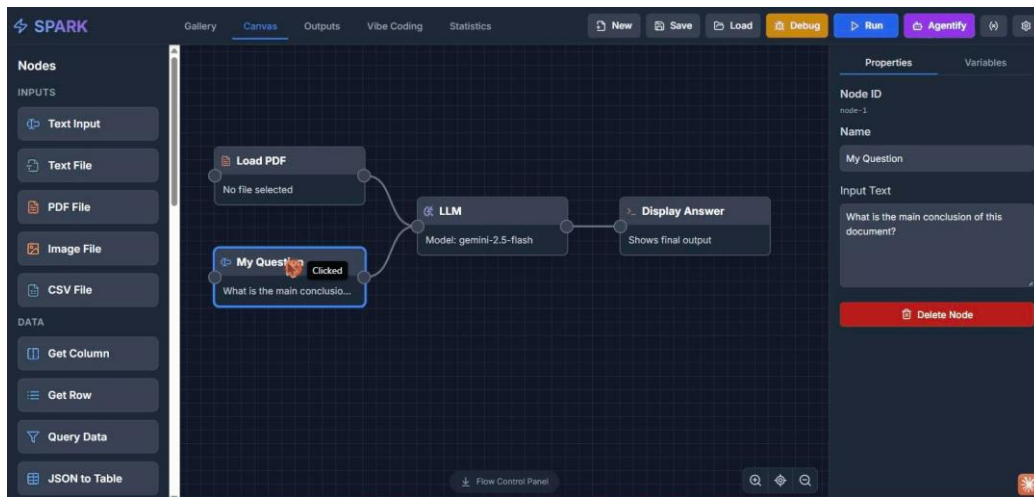


Figure 1 - The PDF Q&A example loaded on the Canvas. We will simplify it into a clean PDF File -> LLM -> Display flow.

Step 3 - Point the LLM at Gemini and write the analysis prompt (Slide 17)

Click the Answer Generator node to open its Properties on the right. This is the LLM node - it decides how the document is analysed.

4. Set AI Provider to Google Gemini. The Model field becomes gemini-2.5-flash.
5. Rename the node to LLM (optional, just for clarity).
6. Replace the Prompt with the analysis instruction below. The {node-0} placeholder pulls in the text of the PDF File node.

Summarise the organisation's sentiment towards the AASB Exposure

Draft ED SR1 (climate-related financial disclosure). Base your answer only on the text below:

{node-0}

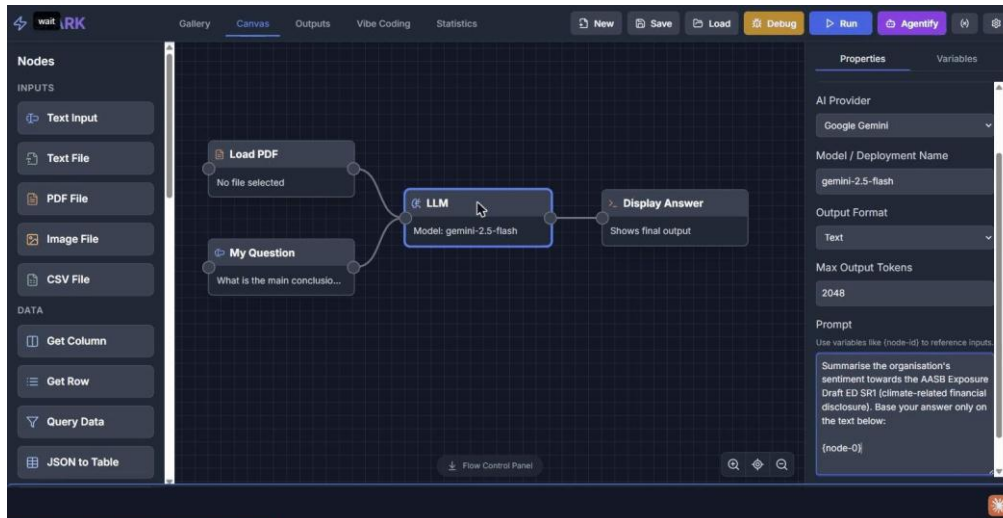


Figure 2 - The LLM node set to Google Gemini / gemini-2.5-flash with the sentiment prompt. {node-0} references the PDF.

Tip: editing the prompt is how you decide what the analysis does - you could just as easily ask for the main concerns, requested changes, or a one-line verdict.

Step 4 - Build the clean flow and upload a letter (Slide 17)

We only need three nodes: PDF File -> LLM -> Display. Select the My Question node and click Delete Node so the LLM reads from the PDF alone.

7. Click the Load PDF node, then Upload PDF File and choose edsr1_nab.pdf.
8. Confirm the node now shows edsr1_nab.pdf and is wired into the LLM, which feeds Display Answer.

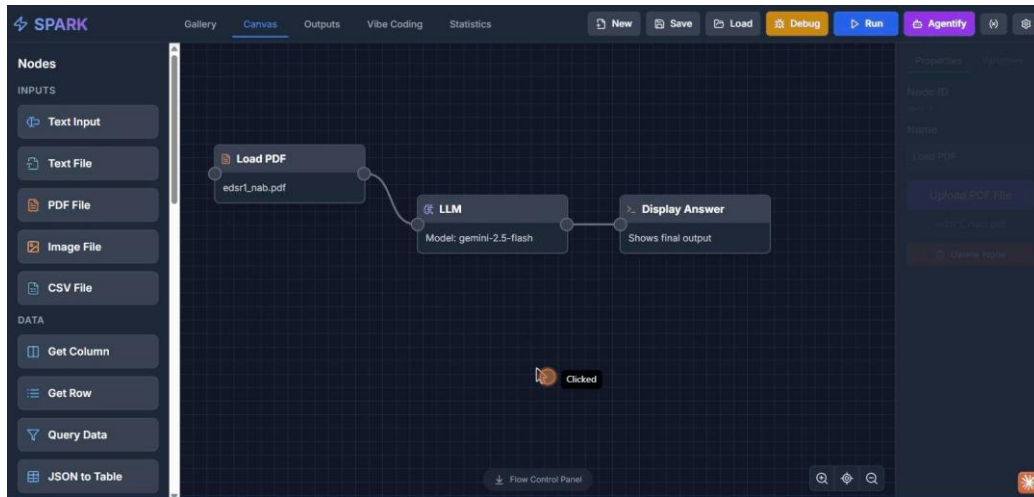


Figure 3 - The finished flow: Load PDF (edsr1_nab.pdf) -> LLM (gemini-2.5-flash) -> Display Answer.

Step 5 - Run the flow and read the sentiment summary (Slide 17)

Click Run (top-right). SPARK reads the PDF, sends the text to Gemini with your prompt, and shows the result in an Agent Output dialog.

For NAB, the model reports a clearly supportive position - for example that the bank is "broadly supportive" of ED SR1 and expects the Standards to improve data availability and quality, while flagging a few areas that need attention.

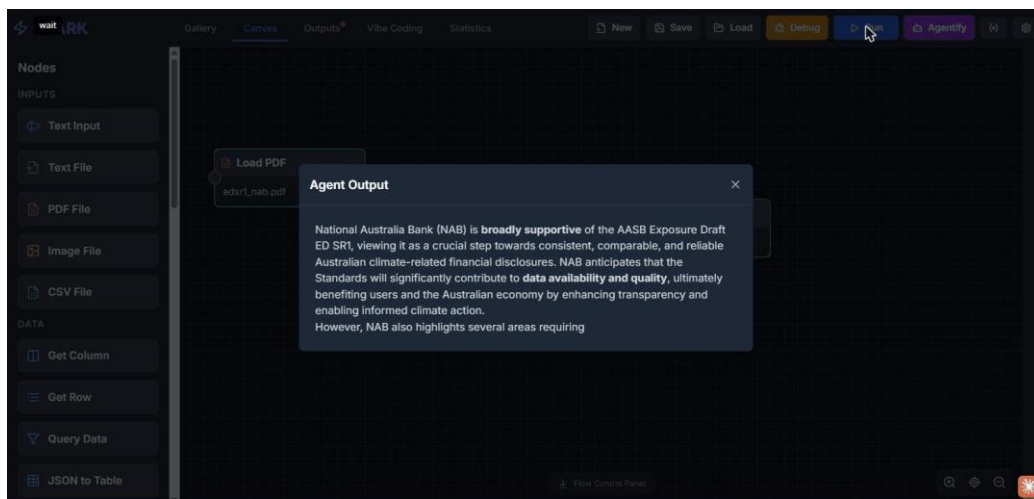


Figure 4 - The Agent Output: a concise sentiment summary of NAB's submission letter.

Step 6 - Reflection (Slide 18)

Before scaling up, pause and judge the output critically:

- Open the letter yourself and skim it. Does the AI summary match what the organisation actually said?

- Is the AI output reliable enough to act on, or only as a first pass that a human still checks?
- How many input tokens were used to read the letter, and how many output tokens were generated? (Check the Statistics tab - longer documents cost more.)

Step 7 - Scale up: handle many letters with an Iterator (Slide 19)

Analysing one letter is useful; analysing every letter is the real win. Replace the single PDF File node with an Iterator that loops over a list of files.

9. Delete the Load PDF node. From the left palette (Logic section) drag an Iterator node onto the canvas.
10. In the Iterator's Properties, set Iteration Type to File List.
11. Click Upload Multiple Files and select all three letters (edsr1_nab.pdf, edsr1_aba.pdf, edsr1_coba.pdf). If the button does not appear, click another node and reselect the Iterator to refresh the panel.
12. Connect the Iterator's output to the LLM's input.
13. Update the LLM prompt so it references the Iterator node instead of the old PDF - change {node-0} to {node-5} (use the Node ID shown at the top of the Iterator's Properties).

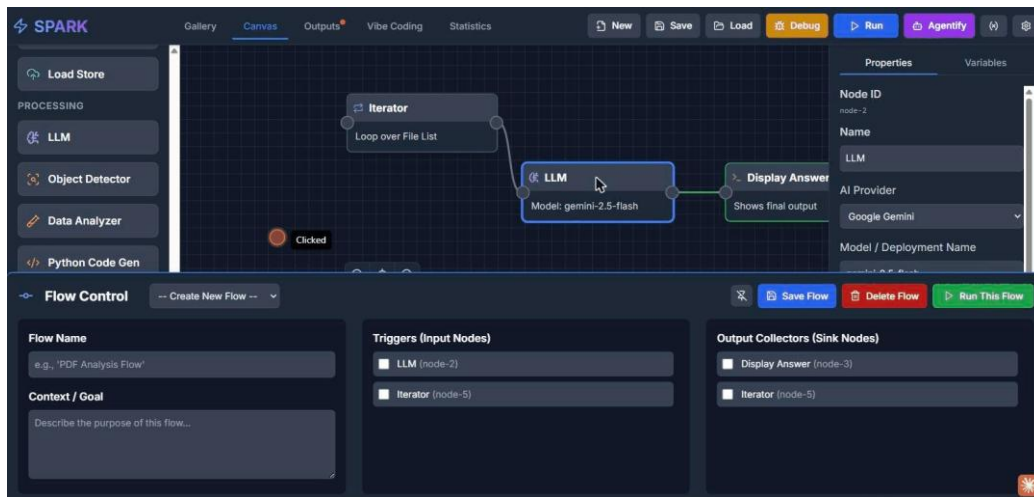


Figure 5 - The Iterator ("Loop over File List") feeding the LLM. Remember to change the node referred to in the prompt.

Tip: the exact Node ID (here node-5) is shown at the top of the Iterator's Properties panel - always copy it from there rather than assuming a number.

Step 8 - Run the batch and compare organisations (Slide 19)

Click Run. The Iterator badge counts up (1/3, 2/3, 3/3) as Gemini processes each letter in turn. Open the Outputs tab to see all results together under one Batch Output.

Now you can compare at a glance: NAB is supportive; the Australian Banking Association (ABA) welcomes the draft and stresses global alignment and comparability with IFRS S1/S2; the Customer Owned Banking Association (COBA) is broadly supportive but raises specific concerns - for example preferring a 1 January 2025 commencement date and flagging the treatment of financed emissions.

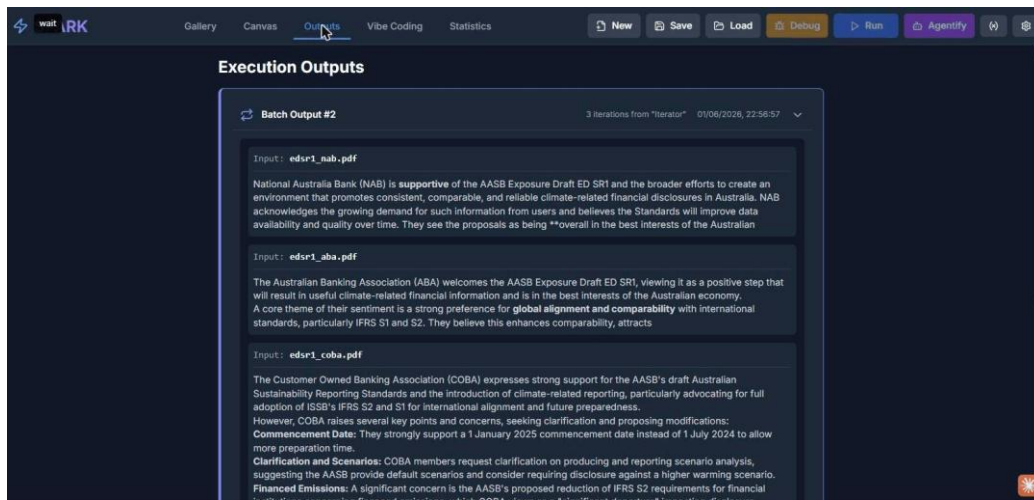


Figure 6 - Batch Output - three iterations from one Iterator, one sentiment summary per letter.

Step 9 - Highlight the evidence: load the Text Highlighter (Slide 20)

A summary is more trustworthy when you can see where it came from. SPARK's PDF Text Highlighter finds the relevant passage with an LLM and highlights it directly in the PDF viewer.

14. Go to the Gallery and find the PDF Text Highlighter card.
15. Click Load Example, then Confirm to clear the canvas.
16. The flow loads with five nodes: My Question and Load PDF feed Find Exact Text, which feeds Combine for Viewer, then View & Highlight.

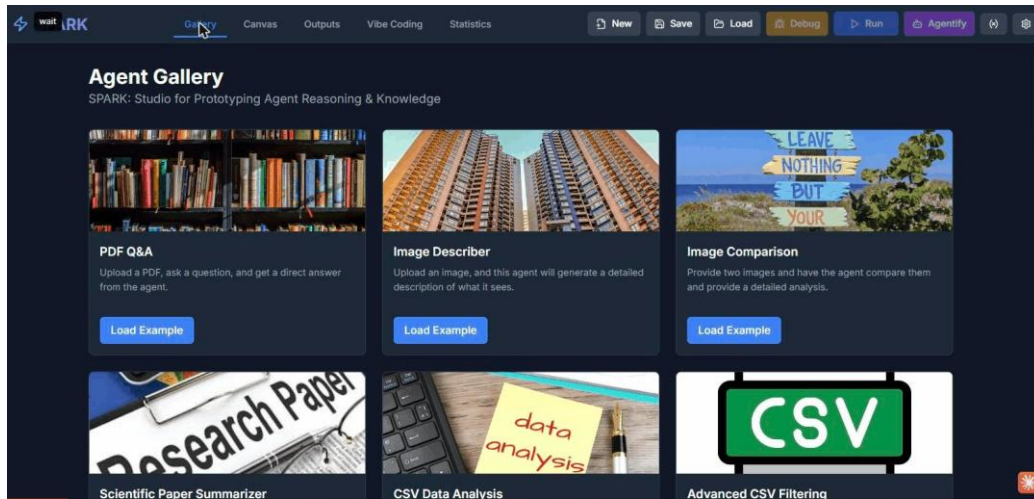


Figure 7 - The PDF Text Highlighter example in the Gallery.

Step 10 - Configure and run the highlighter (Slide 20)

17. Click Find Exact Text and set its AI Provider to Google Gemini (Model becomes gemini-2.5-flash).
18. Click Load PDF, Upload PDF File, and choose edsrl_nab.pdf.
19. Leave the My Question node as "What was the conclusion?" (or write your own question).
20. Click Run and wait for every node to turn green.

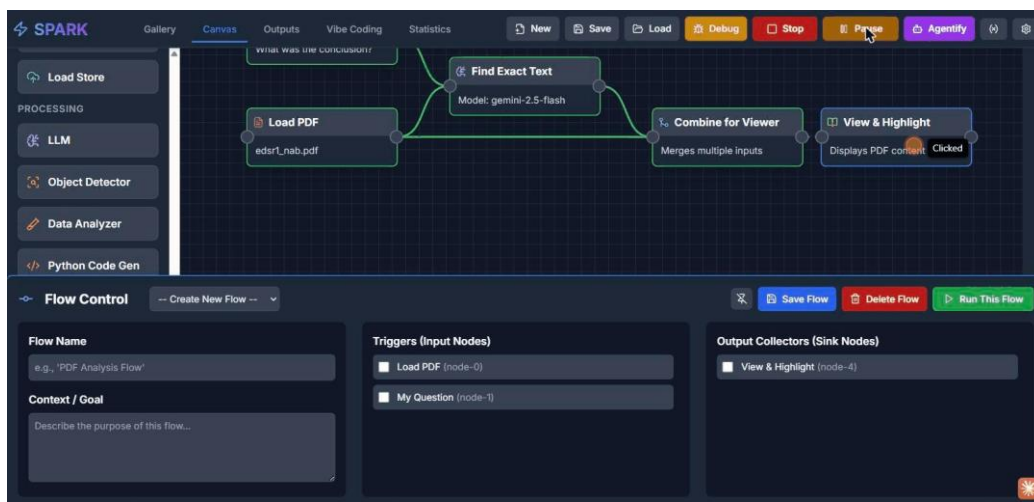


Figure 8 - The highlighter flow: My Question + Load PDF -> Find Exact Text (gemini-2.5-flash) -> Combine for Viewer -> View & Highlight.

Step 11 - View the highlighted PDF (Slide 20)

Click the View & Highlight node to open the PDF viewer. The passages the LLM identified as answering your question are highlighted in place, so you can verify the AI's reasoning against the source document.

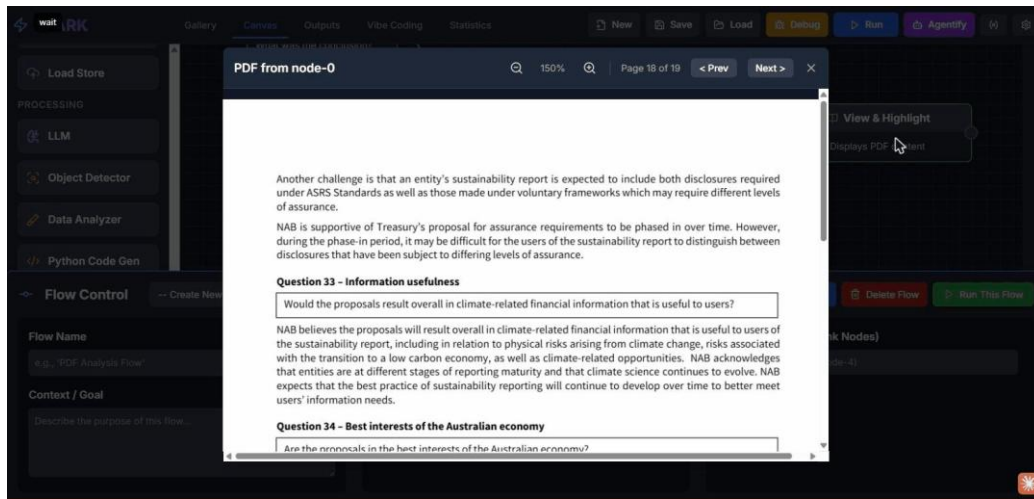


Figure 9 - The NAB letter in the viewer with the relevant text highlighted - the evidence behind the AI's answer.

Tip: text/object highlighting can be a little unstable in the current SPARK version, and quality depends on the model. If a highlight looks off, rerun or refine your question.

Good to know

- Set your Gemini key in Settings before running anything - flows fail silently if the provider has no key. Your key is never shown in screenshots or shared.
- AI output is a fast first pass, not the final word. Always open the source letter (or use the highlighter) to verify before relying on a summary.
- Watch your token use on the Statistics tab - long PDFs consume more input tokens, which is worth knowing when you scale to many documents.
- The Iterator turns a one-document flow into a many-document pipeline just by swapping one node and updating the prompt's node reference.
- If a node's properties look empty or stale, click away and reselect the node to refresh the panel.